3.2 Kolmogorov-Smirnov goodness-of-fit tests

The Kolmogorov-Smirnov (K-S) test can be used to decide if a sample comes from a population with a specific distribution. In this case it is also referred to as the one-sample K-S test. It is a nonparametric test, i.e. it does not depend on the underlying distribution being tested.

The test is based on the empirical cumulative distribution function, $F_n(x)$, of the sample. For a sample of n i.i.d. variables X_i, this is defined as

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} I\{X_i \leq x\}$$

where $I\{\}$ is the indicator function. Given n ordered data points (from smallest to largest) $x_{(1)}, x_{(2)}, \ldots, x_{(n)}$, the function $F_n(x)$ can be written as

$$F_n(x_{(i)}) = \frac{i}{n}$$

i.e. a step function that increases by $1/n$ at the value of each ordered data point.

The Kolmogorov-Smirnov one sample test is generally more efficient that the chi-square tests for goodness of fit for small samples, and it can be used for very small samples where the chi-square test does not apply.

It is based on comparing two cumulative distribution functions (CDFs).

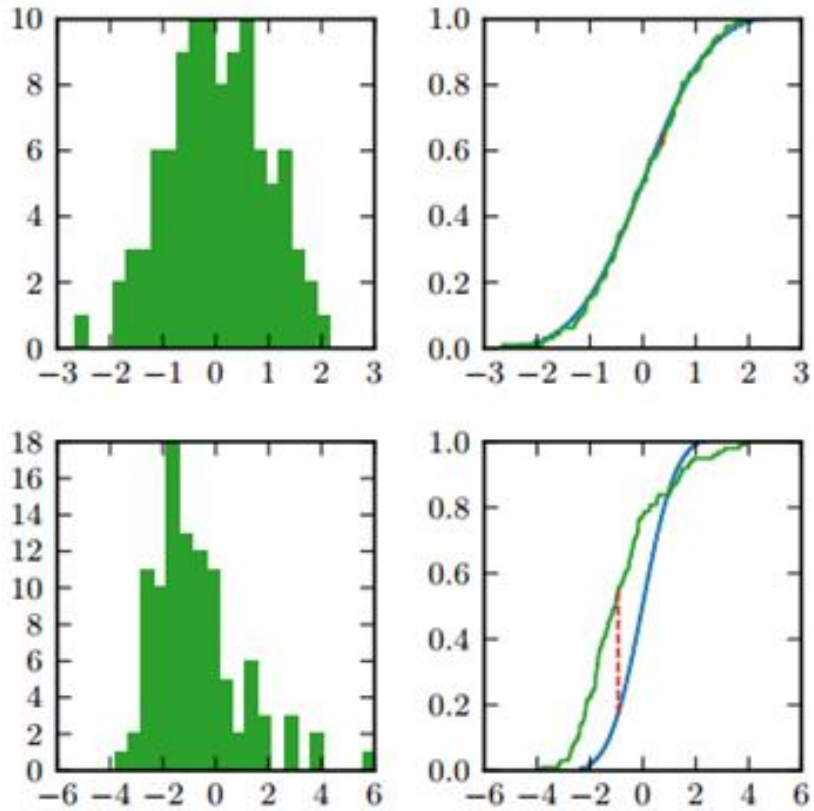


Figure 3.1: Two Kolmogorov-Smirnov test plots (right column) with histograms of the data being tested (left column). On the top row, the empirical CDF (green) matches the test CDF (blue) closely, and the largest difference (dotted vertical red line, near 0.5) is very small. On the bottom, the empirical CDF is quite different from the test CDF, and the largest difference is much larger.

To test H_0: the data follow a specified distribution, against H_1: the data don't follow the specified distribution, we use a test statistic based on the figure:

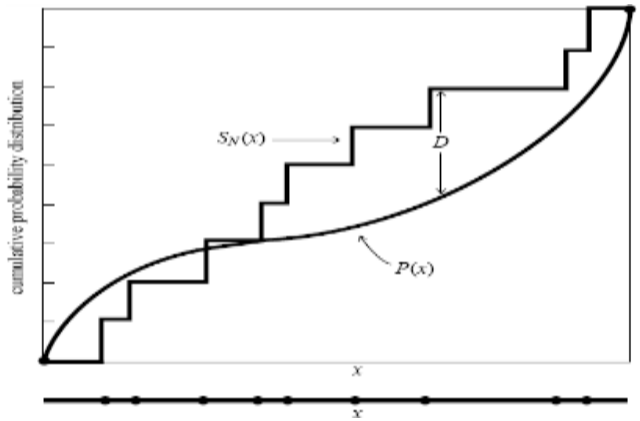


Figure: KS statistic D_n. Compare $P(x) \equiv F(x)$ to $F_n(x) \equiv S_n(x)$. D_n is the greatest distance between the two CDFs.

$$\begin{aligned} D_n &= \sup_x |F_n(x) - F(x)| \\ &= \max_{1 \le i \le n} \left(F(x_i) - \frac{i-1}{n}, \frac{i}{n} - F(x_i) \right) \end{aligned}$$

We reject H_0 at significance level α if $D_n > D_n^\alpha$, where D_n^α is a critical value associated with the sampling distribution of D_n, and is obtained by tables (or use p-value provided in R).

- K-S test (in this form) only applies to continuous distributions that have been fully specified. It is not valid if the parameters of the distribution have been estimated from the data.
- Also, the test is not very sensitive at the tails of the distribution.

Example 3.1

Let us consider a sample of size 10:

0.58, 0.42, 0.52, 0.33, 0.43, 0.23, 0.58, 0.76, 0.53, 0.64

Test the hypothesis that the distribution of the sample is uniform on [0, 1]
i.e. H_0: F(x) = x.

Solutions:

```
> sort(x)
 [1] 0.23 0.33 0.42 0.43 0.52 0.53 0.58 0.58 0.64 0.76
```

H_0: F(x) = x

H_1: F(x) ≠ x

x	Freq	$F_n(x)$	F(x)	$D^- = F(x_i) - F_n(x_{i-1})$	$D^+ = F_n(x_i) - F(x_i)$	$D=\max(D^-, D^+)$
0.23	1	0.1	0.23	0.23 - 0 = 0.23	0.1 - 0.23 = −0.13	0.23
0.33	1	0.2	0.33	0.33- 0.1 = 0.23	0.2 - 0.33 = −0.13	0.23
0.42	1	0.3	0.42	0.42- 0.2 = 0.22	0.3 - 0.42 = −0.12	0.22
0.43	1	0.4	0.43	0.43- 0.3 = 0.13	0.4 - 0.43 = −0.13	0.13
0.52	1	0.5	0.52	0.52- 0.4 = 0.12	0.5 - 0.52 = −0.02	0.12
0.53	1	0.6	0.53	0.53- 0.5 = 0.03	0.6 - 0.53 = 0.07	0.07
0.58	2	0.8	0.58	0.58- 0.6 = −0.02	0.8 - 0.58 = 0.13	0.13
0.64	1	0.9	0.64	0.64- 0.8 = −0.16	0.9 - 0.64 = 0.26	**0.26***
0.76	1	1.0	0.76	0.76- 0.9 = −0.14	1.0 - 0.76 = 0.24	0.24

Since D_n= max(D^-, D^+) = 0.26 < $D_n^{\alpha} = D_{10}^{0.05}$ =0.410, we do not reject the H_0 that the distribution of the sample is uniform on [0, 1] at α=0.05.

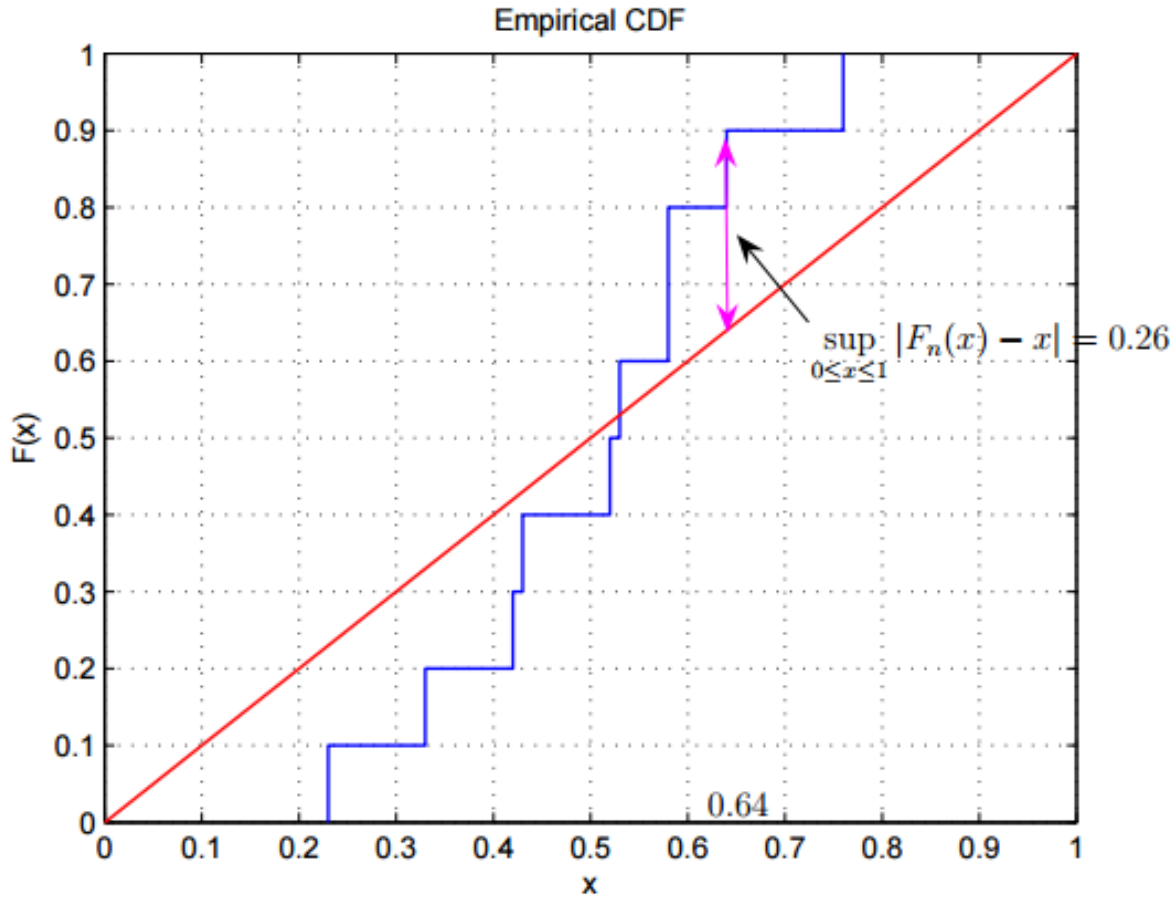


Figure 1: F_n and F_0 in the example.

In R:

```
> ks.test( x, punif, 0, 1)

        One-sample Kolmogorov-Smirnov test

data:  x
D = 0.26, p-value = 0.5085
alternative hypothesis: two-sided

Warning message:
In ks.test(x, punif, 0, 1) :
  ties should not be present for the Kolmogorov-Smirnov test
```

Kolmogorov-Smirnov Table

The table gives the critical values of the D_n as described in Kolmogrov-Smirnov Test.

n \ α	0.01	0.05	0.1	0.15	0.2
1	0.995	0.975	0.950	0.925	0.900
2	0.929	0.842	0.776	0.726	0.684
3	0.828	0.708	0.642	0.597	0.565
4	0.733	0.624	0.564	0.525	0.494
5	0.669	0.565	0.510	0.474	0.446
6	0.618	0.521	0.470	0.436	0.410
7	0.577	0.486	0.438	0.405	0.381
8	0.543	0.457	0.411	0.381	0.358
9	0.514	0.432	0.388	0.360	0.339
10	0.490	0.410	0.368	0.342	0.322
11	0.468	0.391	0.352	0.326	0.307
12	0.450	0.375	0.338	0.313	0.295
13	0.433	0.361	0.325	0.302	0.284
14	0.418	0.349	0.314	0.292	0.274
15	0.404	0.338	0.304	0.283	0.266
16	0.392	0.328	0.295	0.274	0.258
17	0.381	0.318	0.286	0.266	0.250
18	0.371	0.309	0.278	0.259	0.244
19	0.363	0.301	0.272	0.252	0.237
20	0.356	0.294	0.264	0.246	0.231
25	0.320	0.270	0.240	0.220	0.210
30	0.290	0.240	0.220	0.200	0.190
35	0.270	0.230	0.210	0.190	0.180
40	0.250	0.210	0.190	0.180	0.170
45	0.240	0.200	0.180	0.170	0.160
50	0.230	0.190	0.170	0.160	0.150
OVER 50	$\frac{1.63}{\sqrt{n}}$	$\frac{1.36}{\sqrt{n}}$	$\frac{1.22}{\sqrt{n}}$	$\frac{1.14}{\sqrt{n}}$	$\frac{1.07}{\sqrt{n}}$

Example 3.2

Each person in a random sample of n = 10 employees was asked about *X*, the daily time wasted at work doing non-work activities, such as surfing the internet and emailing friends. The resulting data, in minutes, are as follows:

108 112 117 130 111 131 113 113 105 128

Test if these data come from a normal distribution with mean 120 and standard deviation 10.

Solutions:

H_0: The data are normally distributed with mean 120 and standard deviation 10.
H_1: The data are not normally distributed with mean 120 and standard deviation 10.

$$F(x) = P(X \le 105) = P(Z \le \frac{105-120}{10}) = P(Z \le -1.5) = 0.0668$$

x	Freq	$F_n(x)$	F(x)	$D^- = F(x_i) - F_n(x_{i-1})$	$D^+ = F_n(x_i) - F(x_i)$	$D = \max(D^-, D^+)$
105	1	0.1	0.0668	0.0668	0.0332	0.0668
108	1	0.2	0.1151	0.0151	0.0849	0.0849
111	1	0.3	0.1841	−0.0159	0.1159	0.1159
112	1	0.4	0.2119	−0.0881	0.1881	0.1881
113	2	0.6	0.2420	−0.1580	0.3580	**0.3580***
117	1	0.7	0.3821	−0.2179	0.3179	0.3179
128	1	0.8	0.7881	0.0881	0.0119	0.0881
130	1	0.9	0.8413	0.0431	0.0587	0.0587
131	1	1.0	0.8643	−0.0357	0.1357	0.1357

Since D_n=0.3580 < $D_n^\alpha = D_{10}^{0.05}$ =0.410, we do not reject H_0 that data are normally distributed with mean 120 and standard deviation 10 at α=0.05.

In R:

```
> ks.test(x, pnorm,120,10)

        One-sample Kolmogorov-Smirnov test

data:  x
D = 0.35804, p-value = 0.154
alternative hypothesis: two-sided

Warning message:
In ks.test(x, pnorm, 120, 10) :
  ties should not be present for the Kolmogorov-Smirnov test
```

There is a plethora of statistical tests available for checking whether a sample of data comes from a population with a specific distribution.

The Kolmogorov-Smirnov test is one useful such test.

We will consider it in more detail at the practical sessions.